UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	ICT450
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of five (5) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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QUESTION 1 **(20 marks)**

a) Data is a raw fact or a building blocks of information. Identify FOUR (4) examples of data for an online store.

(4 marks)

b) Describe TWO (2) differences between a database and a spreadsheet.

(4 marks)

c) There are three rules of referential integrity which are Insert Rule, Update Rule and Delete Rule. Describe these three rules and illustrate your answer.

(6 marks)

d) Database system is designed based on two design strategies. Explain THREE (3) differences between these two designs.

(6 marks)

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QUESTION 2 **(20 marks)**

a) Briefly describe business rule and state THREE (3) essentials of identifying and documenting business rule to a company.

(8 marks)

b) Explain an importance of index in database.

(2 marks)

c) List FOUR (4) common sources of database failure.

(4 marks)

d) Briefly describe THREE (3) levels of backup that could be performed in database recovery management.

(6 marks)

QUESTION 3 (20 marks)

Table 4 shows the reservation data for a dinner at a Queen Hotel.

Table 4: Reservation data

CustomerID	A01	A01	A02	A03
Reservation ID	R1	R3	R2	R4
Reservation Date	19/02/2022	20/02/2022	19/02/2022	23/02/2022
Time Slot	8:00 PM	8:00 PM	9:00 PM	7:00 PM
First Name	Intan		Fatin	Masitah
Last Name	Nasa		Ruslan	Mat
Email	intan@gmail.com	intan@gmail.com	fatin@gmail.com	mas@gmail.com
Phone Number	017-12345678			012-3455777
Number of Guests	2	3	1	1
Staff In Charge	111 – Yusri	222 – Fashiha	111 – Yusri	333 - Laila
Type Of Cuisine	Italian	Japanese	Italian	Chinese
Menu ID and Name	IT1 – LasagnaIT2 - Pizza	J1 - GyozaJ2 - Sashimi	IT3 - Tiramisu	C1 - Kung PaoChickenC2 - Fried Noodles

Answer the following questions.

a) State the normal form of the above table. Explain your answer with reasons.

(3 marks)

b) Convert the table into a relational schema. Identify the primary key(s) and partial dependencies and/or transitive dependencies.

(6 marks)

c) Normalize the relational schema in (b) until third normal form (3NF). Show all your work.

(11 marks)

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QUESTION 4 **(20 marks)**

The relational schema below shows the movie database system.

```
MOVIE (MovieID, MovieName, Genre, DateRelease, NetProfit, DirectorID*)
DIRECTOR (DirectorID, DirectorName, Nationality)
REVIEW (ReviewID, Description, MovieID*, ReviewerID*)
REVIEWER (ReviewerID, ReviewerName)
```

Write the SQL statements based on the following requirements:

a) Display the movies that have been released from 2021 until 2023.

(4 marks)

b) Identify the total movies that have been directed by Malaysian directors.

(4 marks)

c) Calculate the average net profit for each genre and name the output as "Average Net Profit".

(4 marks)

d) Identify the reviewer's name(s) that has made more than 2 reviews.

(4 marks)

e) Display the movie that received the most reviews.

(4 marks)

QUESTION 5 (20 marks)

Auto Servis Sdn Bhd launched their first ever automotive company. Auto Servis Sdn Bhd plan to hire a database designer to build their own Car Dealership System (CDS). These requirements are required to model the database of CDS :
Each salesperson is managed by one manager and each manager needs to manage many salespersons. Manager has manager ID, first name and last name. A salesperson has their ID, first name and last name. A salesperson is assigned to more than one customer. Each customer has their customer ID, first name, last name, phone number, address, city, state and country. Each customer can buy more than one vehicle. Every vehicle has its own, unique Vehicle Identification Number (VIN) which also referred to as the body number. Each vehicle can be categorized as Car, Truck and SUV. Car has engine size, truck has tonnage and SUV has number of seats as their attribute respectively. An invoice is issued to customers. It includes a unique invoice number, a salesperson's ID, VIN as well as the date of invoice.

Based on the given information :

a) Construct a complete Entity Relationship Diagram (ERD) with all required entities, attributes, relationships as well as the cardinalities using the Crow's Foot notation and include relevant attributes when necessary.

(16 marks)

b) Briefly describe TWO (2) examples of reports that can be generated from ERD in (a).

(4 marks)

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